Regression Analysis of Home Price Factors

By

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Chair		Date

TABLE OF CONTENTS

Abstract

The housing market has been an ever-changing market throughout the years, becoming increasing more complex with every change in the economy. Research has shown that in the long run real estate always seems to increase in value, yet some seem to lag behind the moving averages. If buyers were able to predict the true value of a home, it would make the purchasing decision process much easier. This research uses data from the King County region of Seattle, Washington to determine if a multiple regression model can be created to predict home value. A regression model was found in SAS using the full model, backward selection and forward selection. The models were not proven to be the best estimators of predicting the value of a home given the specific variables. Rather, more research must be done, taking other variables into consideration.

Chapter 1

Introduction

1.1 Introduction and History

Owning real estate and land has always been associated with the assumption of wealth. It first became prevalent after the President Thomas Jefferson made the Louisiana Purchase for $15 million. [1] People rushed west with the hopes of becoming rich and the acquisition of land. Since then the population of America is now ten times what is was in the 1860's. As the population grew and America went through many cultural and economic changes occurred, one thing stayed the same. Ownership of land is still associated with the accumulation of wealth. The price per acre in the Louisiana Purchase approximately 4 cents per acre. On average in the United States the price per acre is over $3100 now, and constantly increasing.

1.2 Statement of Problem

Due to the importance of real estate, our country now has over 2 million active real estate agents. More than 900,000 new houses are projected to be sold this year, up Problem is the amount of land and number of people have a negative relationship. The amount of available land is decreasing as the gross population is growing year over year. People are making purchases of homes on this land without the understanding of true value, causing an asymmetric influx in information. Consumers can observe that the ownership of home is valuable in grand scheme of building wealth, but the difference between a wealth building homeowner and the general homeowner is information.

For this reason, in this study, I will analyze factors that could impact the value and selling price of a home. The data from Kaggle.com will be analyzed using Multiple Regression. After looking at the data and doing some research, I will decide if the price of a home can effectively be predicted based on the variables within the data.

1.3 Relevance of the Problem

More personally, as a soon-to-be graduate, I have become increasingly more interested in purchasing a home. There are many factors that I am taking into account in this process, many of which are within this dataset: living square footage, lot square footage, number of bedrooms, number of bathrooms, year built, etc. To maneuver this process the use of realtors and real estate agents have become the prominent source for purchasing advice on an individual home basis. Working for a home builder has broadened my horizons and amount of information I can obtain from a primary source. As a CAD draftsman for a custom home builder, I am on the other side of the home purchasing process. I am given opportunities constantly to advise and interpret the process with family and friends. As someone who will be providing a secondary source of information, I want to be aware of the factors that play the largest role in selling price of a home to better help those around me.

1.4 Literature Review

When a person enters the realm of possibility of owning a home it is usually the first major purchase and considered the key to adulthood. Whether they are making this purchase as a joint effort with a partner or going through this process alone, determining a reasonable price for a home is a daunting task. Even for a long tenured real estate agent, there are times when their estimates for a home value based on its price are completely wrong.

According to the National Association of Realtors, "The average household income of a first-time buyer has increased to over $74,000 and to over $101,600 for repeat buyers." [2] This number is relevant for the fact that those numbers are 18 percent and 60 percent greater, respectively, from median household income of $63,179. This further emphasizes the need for as much information as possibly attainable for a buyer and especially for a first-time buyer.

Many factors go into the evaluation of home and more specifically the price of which a buyer will pay for it. It is said that if a home is owned by a family for over ten years, they will value the home for 66% more than any price a buyer will look to pay for it. The price of a home can be indicated by factors of importance, not including emotion such as; number of bedrooms, number of bathrooms, square footage of living space, square footage of the lot, condition of the home, year built, and zip code. [5]

Number of Bedrooms

In the real estate world, a space can be considered a bedroom if it has a door that can be closed, a window, and a closet. "The closet requirement is not covered in the IRC and is instead a bedroom feature more related to comfort and livability than safety." [3] For this model we will only consider a room to be a bedroom if there is a closet present.

Number of Bathrooms

The number of bathrooms can be broken down into quarters (1/4), with the four pieces being a sink, a bathtub, a shower, and a toilet. Therefore, a 0.75 bathroom would be understood to be lacking one of the four previous utilities mentioned. For this model we will be following this real estate standard of counting.

Square Footage of Living Space

Square footage of living space is to be considered all "heated living areas" and the interior walls and dead space within that area. It is calculated by multiplying all length and width measurements of all unheated areas. Add all the measurements together to and subtract that number from the total square footage to obtain the available living square footage.

Square Footage of the Lot

Square footage of the lot is the calculation of the land that house is built upon and will be owned by the buyer. The county, and sometimes Homeowner's Association, within which the house is built can place regulations on where the house can be placed on the lot. Examples such as, "Front Setback of 25ft."

meaning nothing can be built within the first twenty-five feet of the front line of the lot.

Condition of the Home

In the real estate world, condition of a home considers all of the intangible features that are never seen by a common buyer. A home inspection is used to evaluate on this scale of a home. These include the qualities of the roof, wiring, plumbing and foundation. For this model we will be using the most standardized version of this scale, on a range of one to five.

Zip Code

Zip code is common term standing for Zone Improvement Plan to allow postal services to group information and deliveries. In the sense of real estate is plays more of geographic-financial role. Many times, zip codes right next to each other will have broadly different average household incomes and home values. In this model zip code will be used exclusively as a reference bearing.

1.5 Limitations

Because of the broad differences between markets across the nation, this study will focus on the King County, Washington housing market. By selecting the use of multiple regression, only those data points with all of the required factors reported were used, resulting in studying and analyzing 21,614 home sales statistics.

Chapter 2

Main Body

2.1 Assumptions of the Model

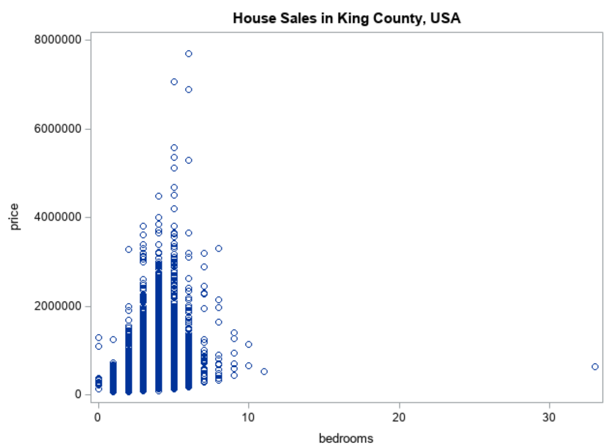
For multiple regression, datasets must be cleaned up by removing outliers. Outliers can skew the data and produce a model that is not the most accurate. Prior to beginning to manipulate the data into a mode, these assumptions must be met. The four pre-requisite assumptions that must be met: linear relationship between the dependent variable and the independent variables, multivariable normality, lack of multicollinearity, and homoscedasticity.

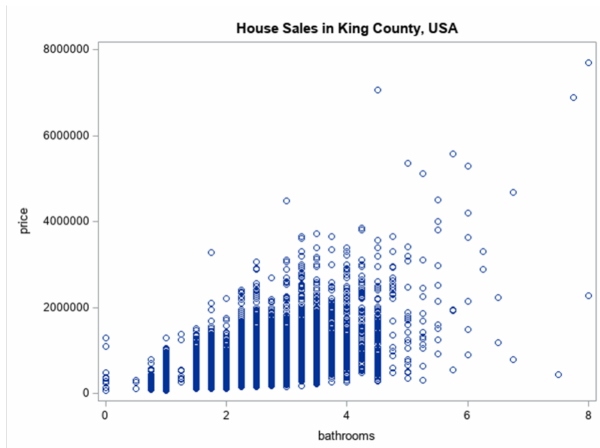
2.1.1 *Data Clean-Up*

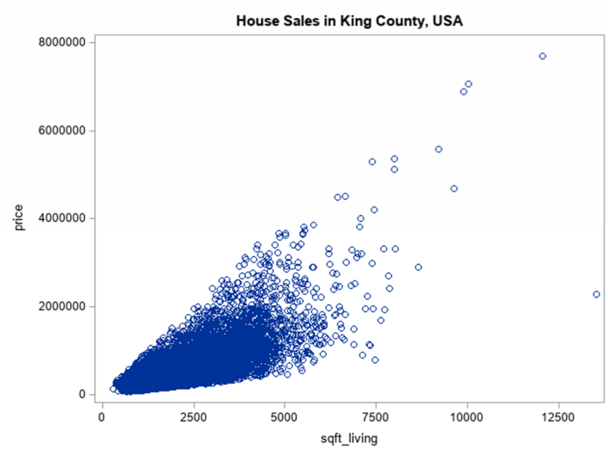
In the process of cleaning up the data, SAS will be used for determining which, if any, data points are classified as an outlier: COVRAT (covariance ratio), R-Student (studentized residual), and Hat Diagonal. When looking at the R-Student calculations, we are looking for outliers with a value greater than 3 or less than -3. The Hat Diagonal cut off for outliers is considered anything greater than $\frac{2p}{n}$ with p parameters and n observations. Lastly the COVRAT measures how much the variability will increase if the data point is dropped. When the Covariance Ratio values are outside the interval of $1 \pm \frac{3p}{n}$ the point should be considered an outlier and possibly deleted from the working dataset.

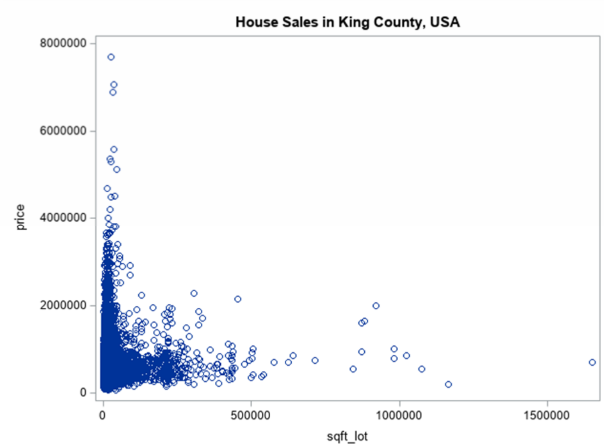
2.1.2 *Linear Relationship*

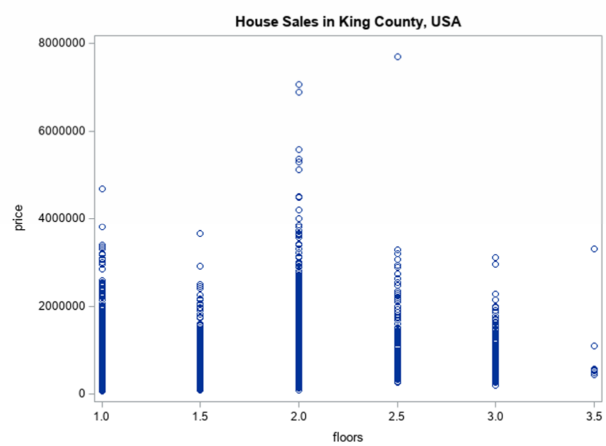
The linear relationship between the dependent variable, price, and each of the independent variables is shown below in the scatterplots. Thus, the linear relationship assumption is met.










House Sales in King County, USA
price
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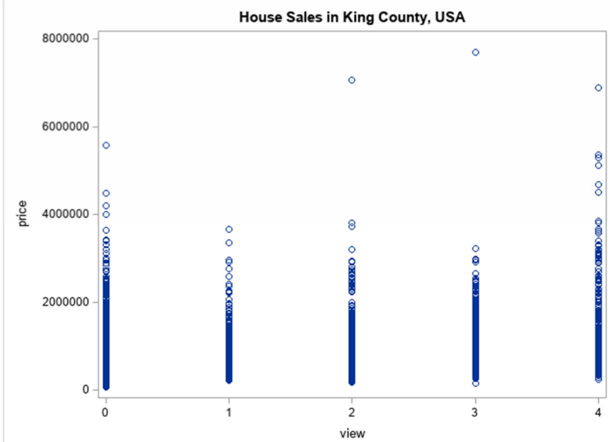
6000000
4000000
2000000
0
1.0
1.5
2.0
2.5
3.0
3.5
floors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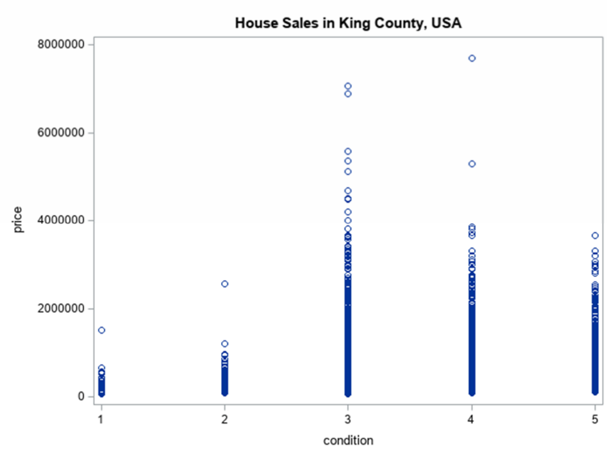
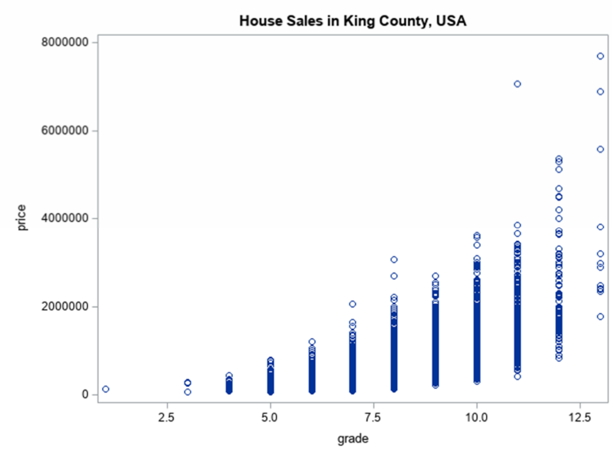
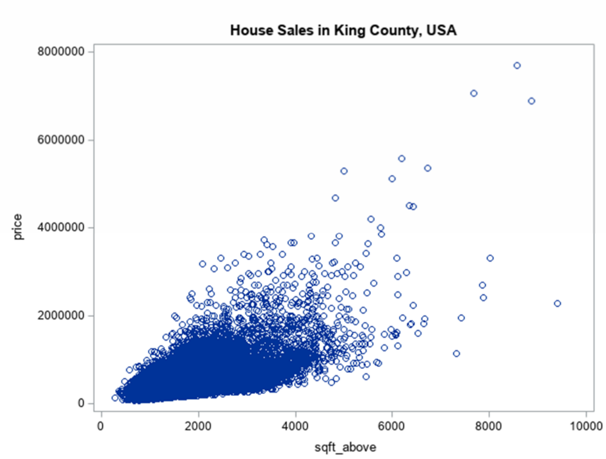
price
8000000
6000000
4000000
2000000
0
0
1
2
3
4
view


House Sales in King County, USA
price
8000000
6000000
4000000
2000000
0
1
2
3
4
5
condition


House Sales in King County, USA
price
8000000
6000000
4000000
2000000
0
2.5
5.0
7.5
10.0
12.5
grade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price
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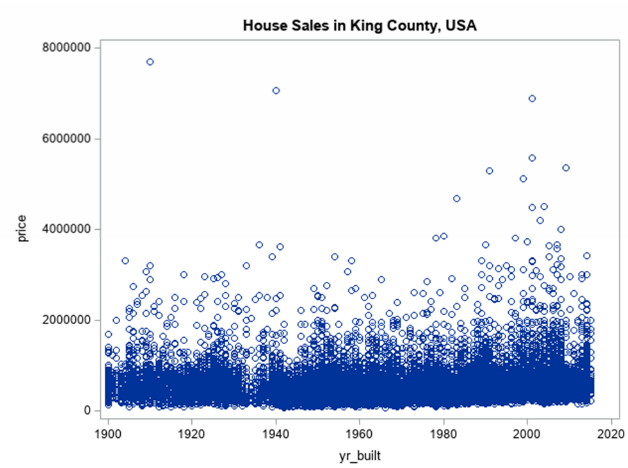
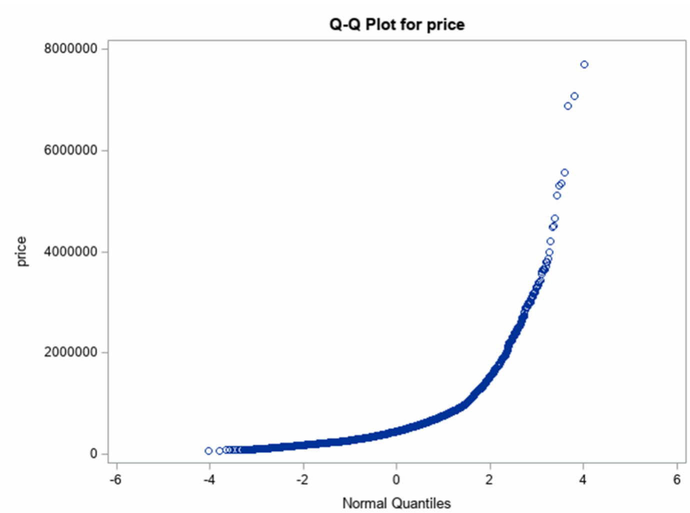
6000000
4000000
2000000
0
0
2000
4000
6000
8000
10000
sqft_above


House Sales in King County, USA
price
8000000
6000000
4000000
2000000
0
1900
1920
1940
1960
1980
2000
2020
yr_built

2.1.3 *Multivariate Normality*

A Q-Q plot will be used to test for normality within the model. From the Q-Q plot we can infer that the data seems to be somewhat normal with a slight skew towards being exponential. For this model of hosing prices and the way they are calculated that is more normal that and it would be for a different dataset. Therefore, this satisfies the assumption of the residuals to be normally distributed.



2.1.4 *Lack of Multicollinearity*

Multicollinearity can be briefly described as occurrence of two or more identified predictor variables in a multiple regression model are highly correlated. Having multicollinearity with model can negatively impact the analysis as a whole and limit the accuracy of the conclusions. There must not be any multicollinearity in the data as this can inflate the variances and covariance yielding an inaccurate model. Variance inflation factor (VIF) will be used to determine multicollinearity. As seen in the tables below, there seemed to be high variance inflation factors for living square footage (sqft_living) and square footage above ground (sqft_above). Due to this, removing square footage above ground will be removed to satisfy the need to eliminate multicollinearity.

Parameter Estimates								
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	Intercept	1	9527998	3089208	3.08	0.0020	.	0
bedrooms	bedrooms	1	-38914	2031.68595	-19.15	<.0001	0.60768	1.64560
bathrooms	bathrooms	1	45706	3496.11415	13.07	<.0001	0.29928	3.34138
sqft_living	sqft_living	1	172.02068	4.61385	37.28	<.0001	0.12083	8.27596
sqft_lot	sqft_lot	1	-0.25889	0.03676	-7.04	<.0001	0.93571	1.06871
floors	floors	1	25153	3809.56752	6.60	<.0001	0.51273	1.95034
waterfront	waterfront	1	573897	18665	30.75	<.0001	0.83203	1.20188
view	view	1	45400	2264.93785	20.04	<.0001	0.72025	1.38841
condition	condition	1	18440	2524.14638	7.31	<.0001	0.80420	1.24347
grade	grade	1	124447	2167.03261	57.43	<.0001	0.33440	2.99044
sqft_above	sqft_above	1	-2.44545	4.52366	-0.54	0.5888	0.15462	6.46732
yr_built	yr_built	1	-3595.88853	74.70655	-48.13	<.0001	0.45059	2.21929
yr_renovated	yr_renovated	1	8.37534	3.92251	2.14	0.0328	0.87403	1.14413
zipcode	zipcode	1	-33.48752	30.98668	-1.08	0.2798	0.78935	1.26686

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	9255732	3047826	3.04	0.0024	0
bedrooms	bedrooms	1	-38901	2031.50455	-19.15	<.0001	1.64536
bathrooms	bathrooms	1	46034	3443.05327	13.37	<.0001	3.24083
sqft_living	sqft_living	1	170.26868	3.28389	51.85	<.0001	4.19259
sqft_lot	sqft_lot	1	-0.26081	0.03659	-7.13	<.0001	1.05869
floors	floors	1	24291	3459.84441	7.02	<.0001	1.60874
waterfront	waterfront	1	573593	18656	30.75	<.0001	1.20079
view	view	1	45578	2241.04016	20.34	<.0001	1.35931
condition	condition	1	18554	2515.18428	7.38	<.0001	1.23470
grade	grade	1	124260	2138.98616	58.09	<.0001	2.91363
yr_built	yr_built	1	-3597.88018	74.61442	-48.22	<.0001	2.21390
yr_renovated	yr_renovated	1	8.38928	3.92236	2.14	0.0325	1.14408
zipcode	zipcode	1	-30.66275	30.54242	-1.00	0.3154	1.23083

2.1.5 *Homoscedasticity*

Homoscedasticity is used to ensure a consistent and similar error variance across all the independent variables. By reviewing the residual plots of each independent variable, a residual plot that is consistent will tell whether homoscedasticity exists within the dataset.

2.2 Formulation of the Model

With all assumptions met, it is now time to work towards finding a model of best fit for the data. We are going to look at some of the tools for diagnosing the data. Consider the following generalized linear model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + E$$

In this generalized linear model where *y* is the dependent variable, x_1, x_2, …, x_k are the independent variables, β_0 represents the *y*-intercept, β_i determines the contribution of the subsequential independent variable x_i for all *i* = 1, 2, …, *k*, and E is the random error variable following normal distribution with mean 0 and standard deviation of σ^2. The least square estimate of βi by $\hat{\beta} i$, *i* = 1, 2, …, *k* and can be jointly computed as:

$$\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)' = (x'x)^{-1}x'y.$$

To formulate the model, all independent and response variables were chosen using industry standards. The dependent variable is the price of the house and the following thirteen independent variables were given in the data and labeled as:

- X_1 represents the number of bedrooms
- X_2 represents the number of bathrooms
- X_3 represents the total living square footage
- X_4 represents the total lot square footage
- X_5 represents the number of floors
- X_6 represents the relative closeness to water

- X_7 represents the quality of the view
- X_8 represents the condition of the home
- X_9 represents the grade of the home
- X_{10} represents the total above ground square footage
- X_{11} represents the year built
- X_{12} represents the year renovated
- X_{13} represents the zip code

2.3 Testing and Analysis

The analysis of variance of a regression model partitions the total sum of squares, SS_T, as $SS_T = SS_R + SS_E$, where SS_R is the sum of squares due to regression and SS_E is the error sum of squares. These sums of are computed as follows:

- $SST = \sum_{i=1}^{n} y_i^2 - {(\sum_{i=1}^{n} y_i)^2}/{n}$
- $SSR = \beta' X' Y - {(\sum_{i=1}^{n} y_i)^2}/{n}$
- $SSE = SST - SSR$

After computing these values, the analysis of variance table to test the utility of the fitted model is given below:

Source	Degrees of Freedom	Sum of Squares	Mean Square	F-statistic
Regression	k	SSR	$MS_R = \frac{SS_R}{df_R}$	$F = \frac{MS_R}{MS_E}$
Error	$n-k-1$	SSE	$MS_E = \frac{SS_E}{df_E}$	
Total	$n-1$	SST		

To determine the best subset, we will look at a few of the different criteria, based on the analysis of variance.

- Adjusted R-square Criterion

  R-square is a percentage of the response variable variation that is explained by a linear model. The objective of the R_a^2 criterion is to find a subset model so that adding more variables to the subset will result only small increases in R_a^2. The larger the R-square, the more the model explains the variability of the model and the less variability attributed to error. Adjusted R^2 is calculated by $1-\frac{n-1}{n-(k+1)}(1-R^2)$. Therefore, the higher the R-square value, the better the model.

- Mallow's C_p Criterion

  Mallow's C_p is related to the mean square error of a fitted value, calculated using

  $$C_p = \frac{SSE_p}{MSE_k} + 2(p+1) - n$$

  - where n is the sample size
  - p is the number of independent variables in the subset model
  - k is the number of independent variables in the full model
  - SSE_p is the sum of squares error of the subset model
  - MSE_k is the mean square error of the full model

  Models with little bias will have values that are less than C_p. Mallow's C_p selects the best subset model with a small value of C_p and one with a value of $C_p \approx p + 1$.

- Mean Square Error (*MSE*) Criterion

Mean square error is simply the estimator for σ^2 of the model. Since a model with less variance is desired, a smaller *MSE* implies less error and the best regression model possible from the dataset.

- Simplicity of Model

If two models are very similar in the diagnostic tools, then the simpler model will be used as the choice because it leaves less room for error.

Model	$\mathbf{R^2}$	C_p	**Adjusted** $\mathbf{R^2}$
Full Model	0.6997	18.0000	0.6995
Removal of Duplicates	0.6523	13.0000	0.6521
Simplified Model	0.6415	12.0079	0.6414

Using the information from the table and the criteria, we find:

Removal of Duplicates: $\hat{y} = 6199005 - 38793x_1 + 45925x_2 + 170.51104x_3 - 0.25712x_4 + 23823x_5 + 573808x_6 + 45335x_7 + 18867x_8 + 124262x_9 - 3573.35136x_{11} + 8.58039x_{12}$

Simplified Model: $\hat{y} = 5589005 - 16429115x_3 + 595802x_6 + 48322x_7 - 135520x_9 - 3269.44945x_{11}$

Chapter 3

Conclusions

3.1 Summary and Interpretations

When analyzing the regression model, we find living square footage, relative closeness to water, quality of the view, grade of the home and the year it was built to be significant contributing factors in determining the price of a home. With waterfront and living square footage being the strongest parameters, the model suggests that homes closer water with larger conditioned living square footage also are bought for a higher price. With number of bedrooms, number of bathrooms, number of floors, square footage of the lot and zip code having less significant influence, they do still contribute to the price of a home.

3.2 Suggestions for Further Study

After determining the best fitted model, it would be natural to construct confidence intervals of different coefficients. Further tests could also be run to determining the significance of the each of the new coefficients.

When I began my study, I selected only a few of the different possible options that were available as factors that impact the price of a home. In further studies, other factors could be impacted or play a more significant role. Factors such as location, whether it be relative distance to good schooling, shopping options or recreational locations. Factors involving the neighborhoods in which the homes are in, average age, household income or number of children in the house. Factors on the buyer side could also be tested for significance: age of the buyer, occupation and marital status.

In my study, I looked only at a densely populated county in Washington, King County. The most populous county in Washington and listed as the 12th-most populous counties in the United States. [4] Studies could be done to investigate factors that impact home price of rural, suburban, and even differing urban communities.

The data used in this study is from the 2017 fiscal year. Further studies could be used to see if the determining factors that most impact the price of a home change over time. A longitudinal study could be used to determine which factors impact the home prices over time; this study could be used as a predictor model in determining home value change over time.

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